size_balanced_max

equity strategy | family: equity-max | status: research | derived from `size_balanced`

What this strategy does

size_balanced with bigger bets for more return: base unit 0.009, 6x leverage cap

Why it should work. PEAD is stronger in small caps; balancing by size keeps the edge but removes a size bet.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.009
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	size balanced max	market (CRSP VW)
Annualized return	8.90%	9.94%
Annualized volatility	13.00%	19.35%
Sharpe (NAV-pct)	0.72	0.59
Sortino	1.02	
Max drawdown	-43.6%	-55.5%
Calmar	0.20	
Final NAV ($10M start)	$52,667,008	
Worst day	-7.15%	
% positive days	49.8%	
Sharpe 1996-2001	0.82	

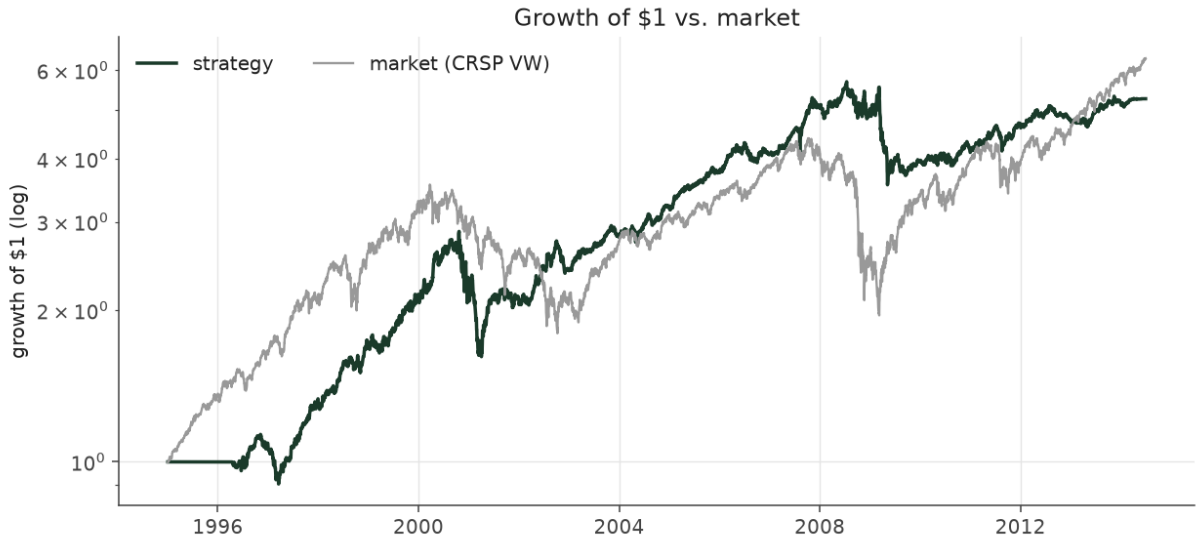
Sharpe 2002-2007	1.52	
Sharpe 2008-2013	0.09	

Correlation of daily returns with the market: **0.04**.

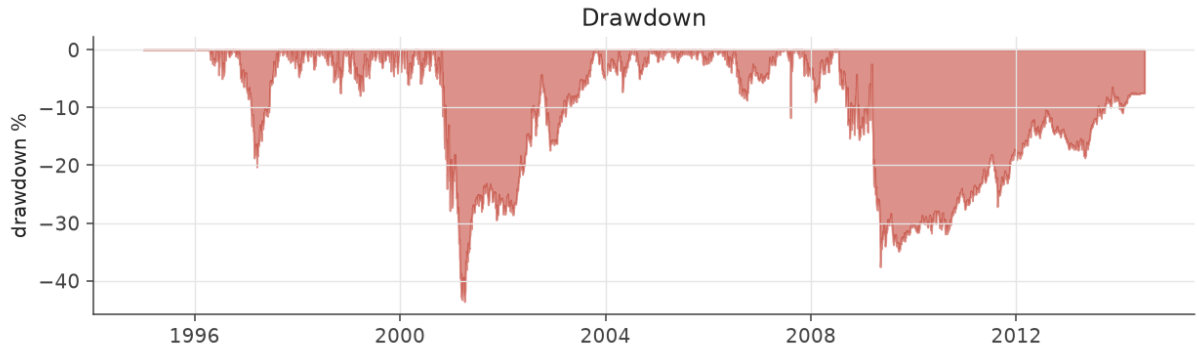
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.721

• best return among equity: `market_boost_max` (0.116); this one: 0.089

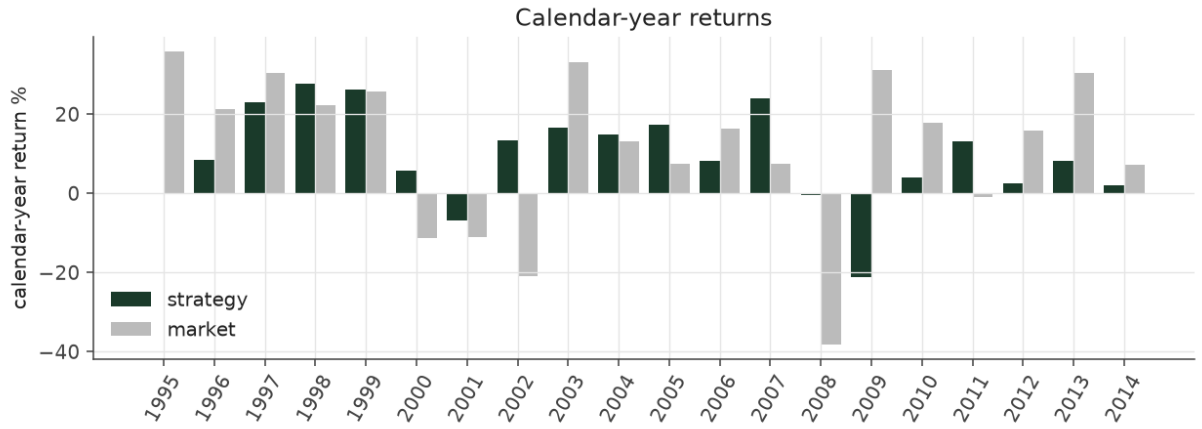
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.204



NAV (log scale)



Drawdown from running peak

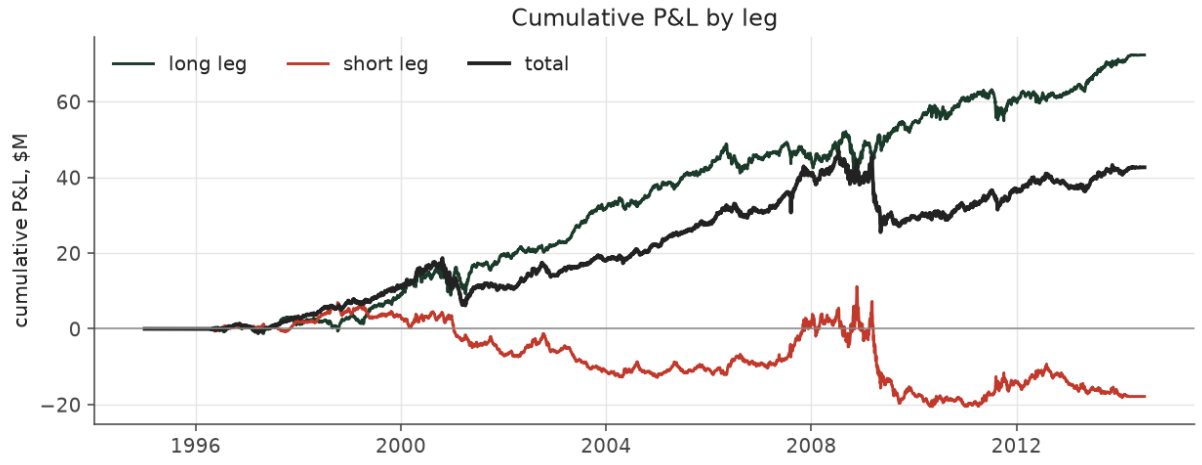


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	72.35	-17.86
P&L after costs ($M)	66.58	-23.92
of which common (average-event) component ($M)	54.98	-55.35

of which excess / PEAD component ($M)	17.37	37.49
excess minus costs ($M)	11.60	31.43
gross return per $ traded (bp)	202.09	-49.26
positions in the money (hit rate)	0.57	0.49
stand-alone Sharpe (leg only)	0.91	-0.19

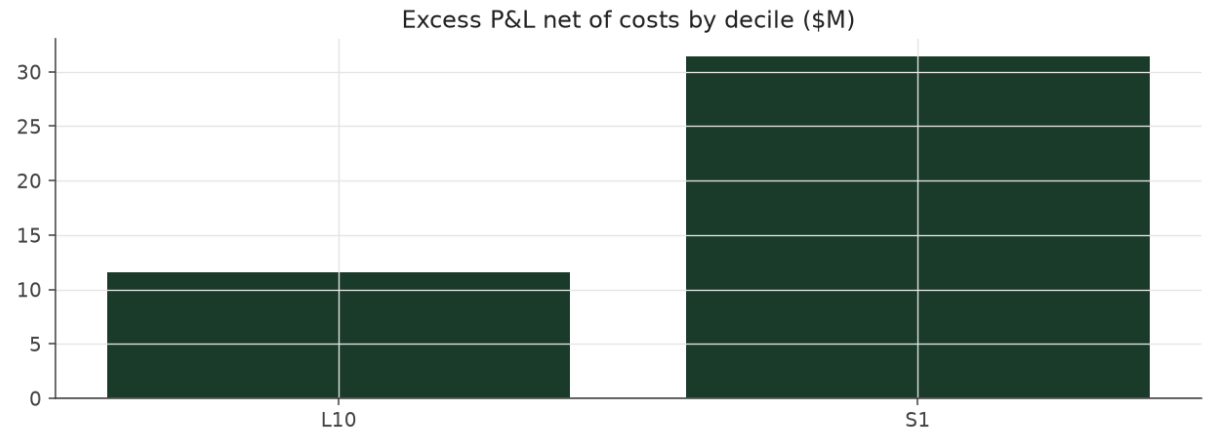
Daily P&L correlation between the two legs: **-0.60**.

• **The short leg loses money in absolute terms** (-17.9M before costs) while the long leg earns +72.4M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 55.0M / 55.3M. Stripping that common component, the surprise signal itself earned +17.4M on the long side and +37.5M on the short side.

• The two legs' daily P&L are -0.60 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.91 long / -0.19 short vs 0.72 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	20,136	3,580.12	66.58	17.37	5.77	11.60	27.0%	32.40
short	1	18,299	3,625.91	-23.92	37.48	6.05	31.43	73.0%	86.68



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	5,287	242.20	-4.46	-2.52	0.78	-3.29	-7.7%	-136.03
2	7,872	960.78	10.40	13.94	2.29	11.65	27.1%	121.25
3	8,605	1,714.13	18.98	21.21	3.23	17.98	41.8%	104.91
4	8,744	2,082.80	9.44	11.76	3.10	8.66	20.1%	41.56
5	7,927	2,206.13	8.31	10.46	2.42	8.04	18.7%	36.43

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $

1	4,793	105.11	2.20	3.07	0.53	2.54	5.9%	242.12
2	7,905	695.75	2.26	5.59	2.09	3.50	8.1%	50.36
3	8,745	1,876.49	20.78	25.47	3.75	21.71	50.5%	115.71
4	8,837	2,320.37	3.77	7.19	3.25	3.94	9.2%	16.97
5	8,153	2,207.82	13.76	13.64	2.21	11.43	26.6%	51.78

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	7,482	1,735.24	3.87	28.13	2.70	25.43	59.1%	146.55
41-60d	11,345	1,880.76	2.18	8.33	3.32	5.01	11.6%	26.64
61-90d	17,258	3,304.06	25.59	9.85	5.18	4.66	10.8%	14.12
>90d	2,350	285.98	11.03	8.55	0.62	7.93	18.4%	277.14

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	10,157	1,151.50	11.16	12.37	1.82	10.55	24.5%	91.60
2002-07	13,612	2,526.59	27.84	30.31	4.13	26.18	60.8%	103.62
2008-13	14,666	3,527.95	3.66	12.18	5.87	6.30	14.6%	17.86

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	7,541	1,314.18	6.42	2.23	2.13	0.10	0.2%	0.77
Chems	776	160.77	1.24	1.77	0.26	1.52	3.5%	94.31
Durbl	990	163.59	-0.74	0.28	0.28	-0.01	-0.0%	-0.60
Enrgy	1,529	318.94	6.24	7.31	0.51	6.79	15.8%	213.03
Hlth	3,443	592.72	4.89	3.10	1.01	2.09	4.9%	35.33
Manuf	4,089	762.89	0.88	3.98	1.23	2.75	6.4%	36.07
Money	5,981	1,218.29	3.03	10.99	1.97	9.02	21.0%	74.06
NoDur	1,707	331.01	2.58	2.92	0.54	2.38	5.5%	71.91
Other	5,674	1,008.37	9.68	10.91	1.78	9.14	21.2%	90.61
Shops	3,935	732.24	6.96	7.77	1.20	6.58	15.3%	89.79
Telcm	1,293	252.06	1.97	2.99	0.39	2.60	6.0%	103.10
Utils	1,477	350.99	-0.48	0.60	0.54	0.07	0.2%	1.97

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	12,942	2,617.72	3.20	9.22	3.98	5.24	12.2%	20.00

low vol	12,792	2,201.21	20.35	23.65	3.90	19.75	45.9%	89.70
mid vol	12,052	2,286.11	18.34	21.21	3.76	17.45	40.5%	76.33

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.90%	-0.00%
0.5%	3.6	8.51%	0.40%
1.0%	7.3	8.08%	0.83%
2.0%	14.6	7.11%	1.79%
5.0%	36.4	2.53%	6.37%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.